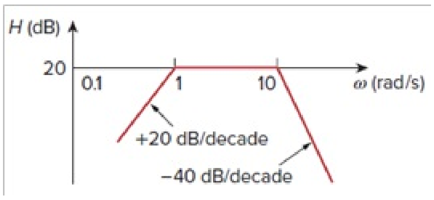
Problem

The Bode magnitude plot of **H**(ω) is shown in Fig. 14.75. Find **H**(ω).

Figure 14.75:



Step-by-step solution

Step 1 of 2

Refer to Figure 14.75 in the textbook.

The initial slope of the plot is $+20\text{ dB/decade}$. Therefore, the transfer function contains one simple zero at origin.

The straight line is shifted by 20 dB indicates that there is 20 dB gain.

$$20\log_{10} K = 20$$
$$\log_{10} K = 1$$
$$K = 10^1$$
$$K = 10$$

At $\omega = 1\text{ rad/sec}$, the slope of the plot is 0 dB/decade, and the change in slope is -20 dB/decade. Therefore, one pole is added to the transfer function at $\omega = 1\text{ rad/sec}$.

The pole at $\omega = 1\text{ rad/s}$ is represented as $\dfrac{1}{\left(1+\dfrac{j\omega}{1}\right)}$.

Step 2 of 2

At $\omega = 10 \text{ rad/sec}$, the slope of the plot is -40 dB/decade, and the change in slope is -40 dB/decade. Therefore, two poles are added to the transfer function at $\omega = 10 \text{ rad/sec}$ and it continues to next frequency at $\omega = \infty \text{ rad/sec}$.

The pole at $\omega = 10 \text{ rad/s}$ is represented as $\dfrac{1}{\left(1+\dfrac{j\omega}{10}\right)^2}$.

The overall transfer function is,

$$\mathbf{H}(\omega) = \frac{K(j\omega)}{\left(1+\dfrac{j\omega}{1}\right)\left(1+\dfrac{j\omega}{10}\right)^2}$$

Substitute 10 for *K*.

$$\begin{aligned}\mathbf{H}(\omega) &= \frac{(10)(j\omega)}{\left(1+\dfrac{j\omega}{1}\right)\left(1+\dfrac{j\omega}{10}\right)^2} \\ &= \frac{10j\omega}{\dfrac{1}{100}\left[(1+j\omega)(10+j\omega)^2\right]} \\ &= \frac{1000j\omega}{\left[(1+j\omega)(10+j\omega)^2\right]}\end{aligned}$$

Therefore, the transfer function of the bode plot is $\boxed{\dfrac{1000j\omega}{\left[(1+j\omega)(10+j\omega)^2\right]}}$.